

MACHINE LEARNING BASED PREDICTION ON CHILD VACCINATION IN INDIA

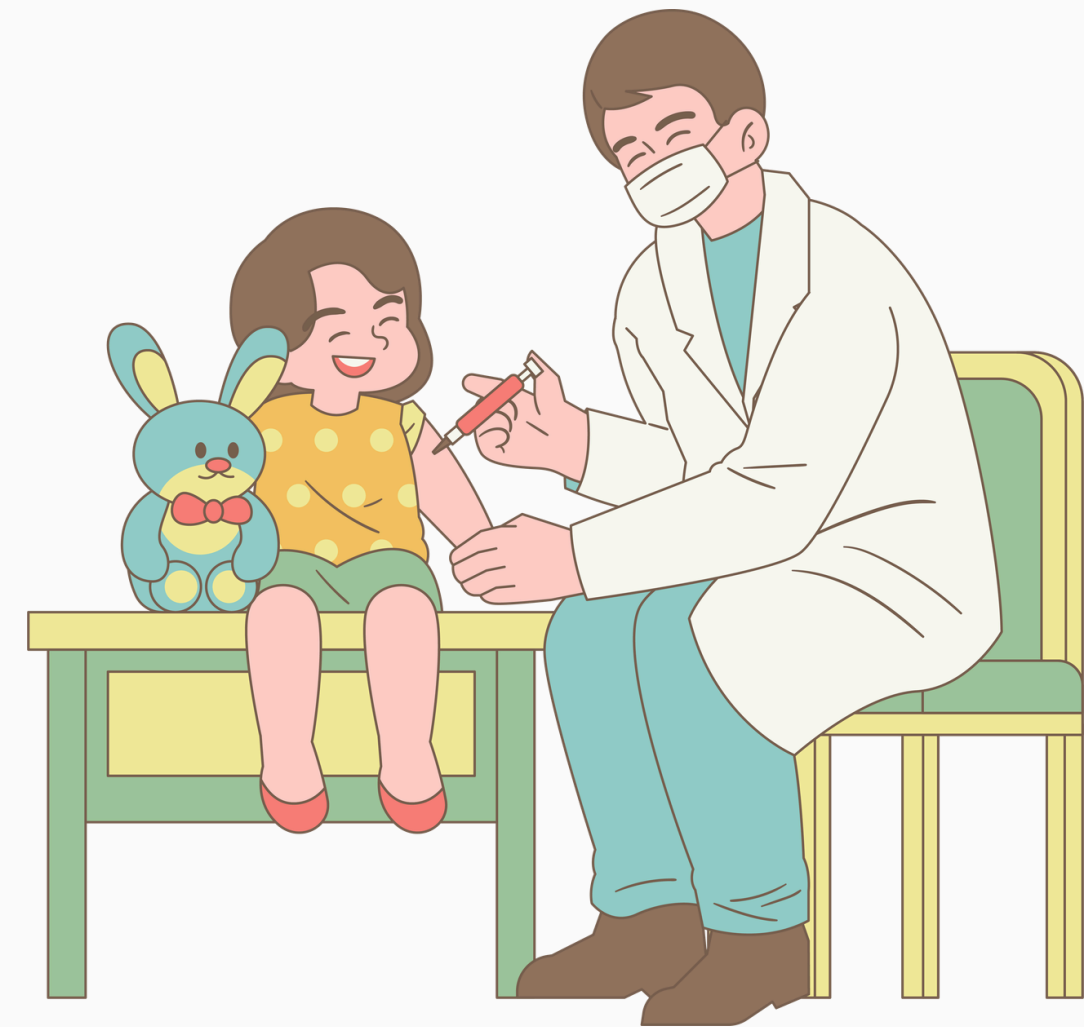
Evidence from NFHS-5

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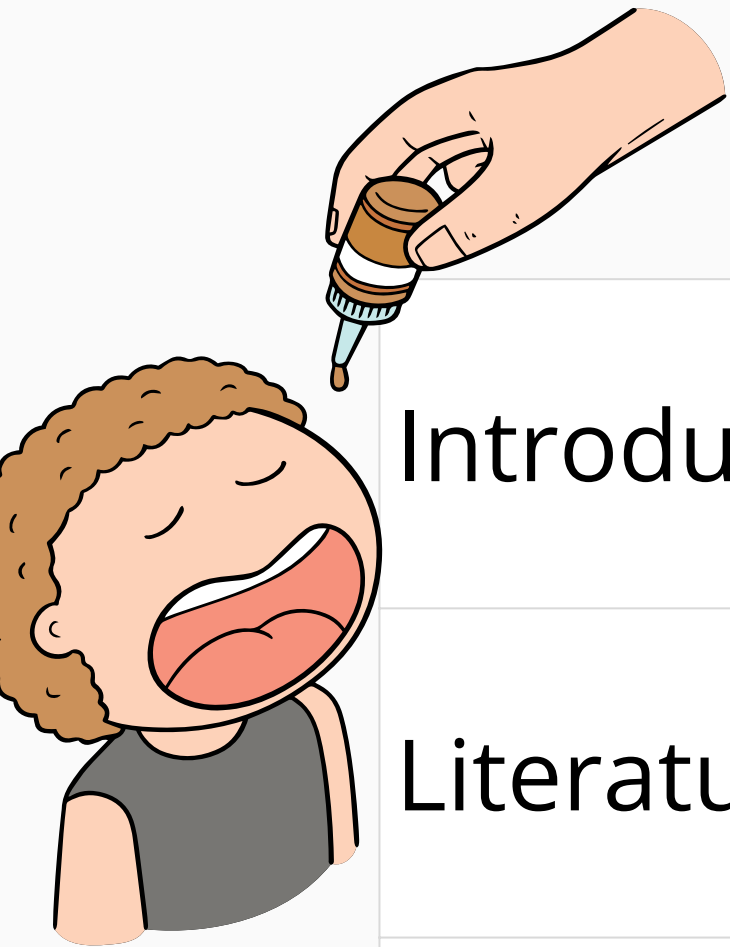


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Childhood Vaccination in India

- Vaccination is one of the most effective and low-cost public health interventions.
- Yet, ~37% of children (12–23 months) in India remain incompletely immunized.
- Gaps persist due to socioeconomic inequality, geographic barriers, and health-system challenges.
- NHM definition: A child is fully immunized if they receive all vaccines scheduled in the first year of life.

Vaccines Required for Full Immunization

- BCG
- 3 doses of DPT / Pentavalent
- 3 doses of Polio (excluding birth dose)
- 1 dose of Measles / MR

Literature Gap & Motivation

- In previous study context, while international studies have successfully used machine learning to predict childhood vaccination, Indian studies have lagged behind.
- Previous Indian vaccination studies relied almost entirely on traditional logistic regression.
- No published study has compared advanced models like SVM, KNN, and XGBoost specifically using the latest NFHS-5 dataset.
- Previous studies lacked comprehensive Exploratory Data Analysis (EDA) on specific vaccines and did not systematically address class imbalance in the data.

Study Objectives

- This study analyzes NFHS-5 data to explore childhood vaccination patterns, key influencing factors
- The findings aim to guide evidence-based policies for improving immunization coverage and equity.
- Multiple machine learning models are trained and compared to predict full vaccination status



Data & Variables Overview

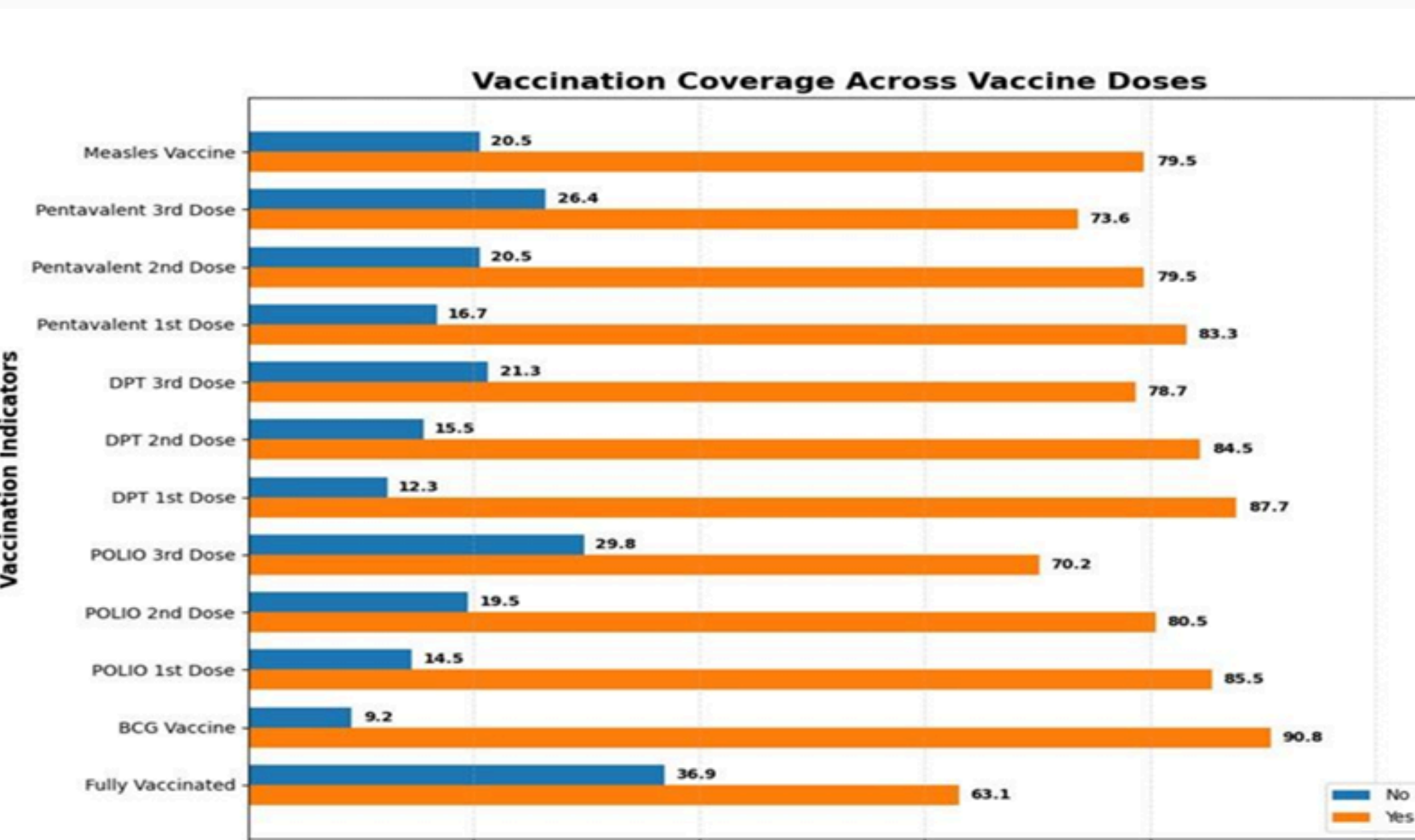
Data is sourced from NFHS-5 (2019–21), accessed via the DHS Program, representing nationally representative child health and immunization information for India.

Sample: 12,377 children aged 12–23 months (filtered from 42,955 records).

Outcome Variable: Childhood vaccination status — Binary (Fully Vaccinated vs. Not) and Multiclass (5 categories by dose completion).

Domain	Variables
Socio-Demographic	Child age, Birth order, Mother's education, Mother's age >35, Residence, Media exposure
Socioeconomic	Wealth index, Automobile ownership, ICDS benefits, Facility distance
Healthcare Access	ANC visits (≥ 4), Place of delivery, Health worker visit

EXPLORATORY DATA ANALYSIS



Descriptive Statistics

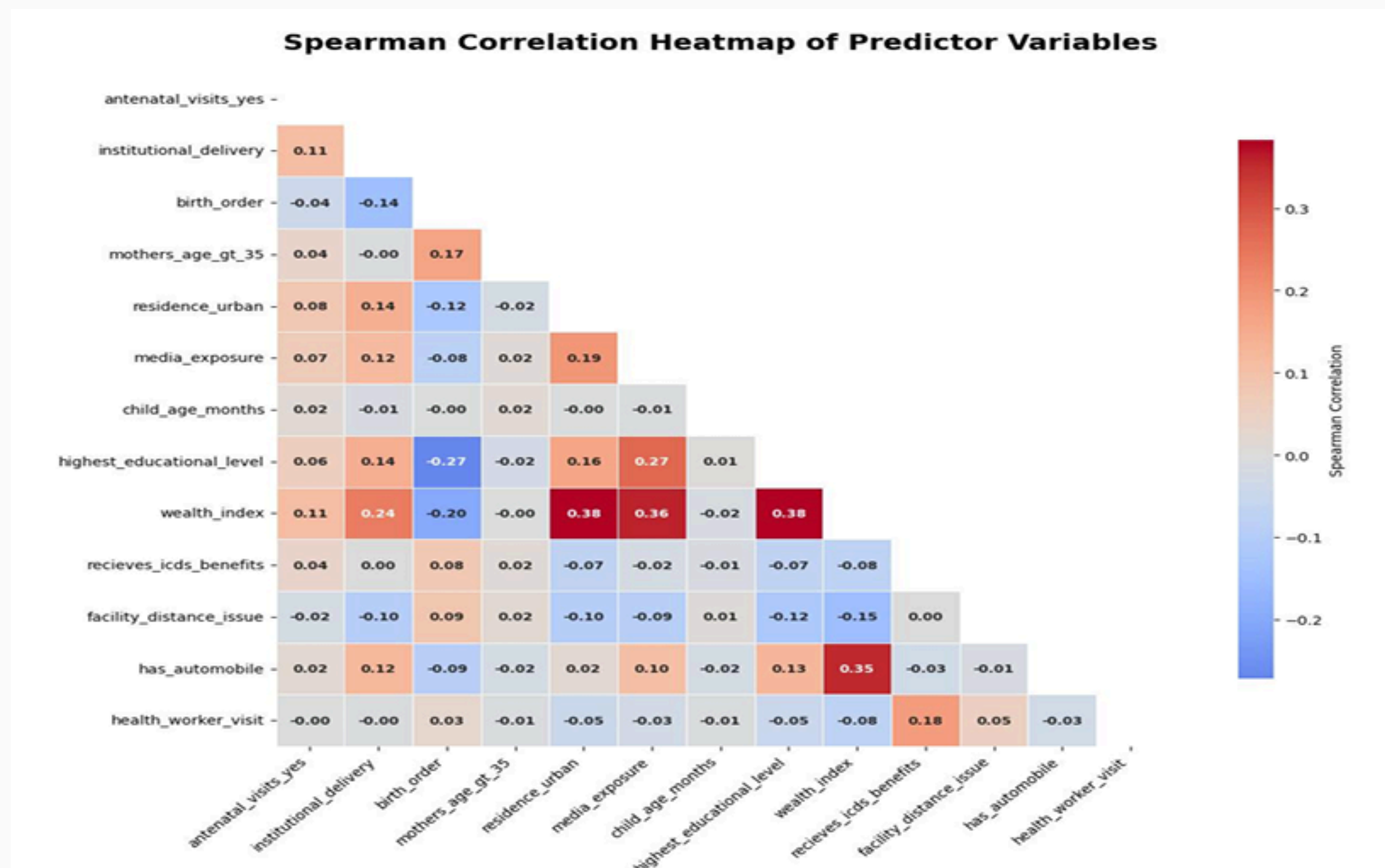
- **Vaccination status:** 63.1% fully vaccinated, 36.9% not vaccinated.
- Frequency distributions summarized for all 13 predictor variables.

Multicollinearity Check (VIF)

- VIF values acceptable for most predictors; child age showed the highest VIF (12.13).

Chi-Square Associations

- All 13 predictors showed significant association with vaccination status ($p < 0.001$), supporting their inclusion in the model.



Correlation Analysis
Spearman correlations among predictors were low ($|r| < 0.40$), indicating no major multicollinearity.

Machine learning Algorithms

Logistic Regression

- Baseline classification model
- Estimates regression coefficients
- Calculates probability for new observations
- Uses threshold (usually 0.5) for classification
- Ridge logistic regression with $\lambda = 0.1$

K-Nearest Neighbors

- Classifies by majority vote of K nearest neighbors (K=11, Euclidean distance). Captures non-linear patterns without distributional assumptions.
- Standard scaling is a preprocessing step where all numerical features are transformed to have: Mean=0, Std deviation=1

Support Vector Machine

- Finds the best boundary (hyperplane) separating vaccinated and non-vaccinated individuals. SVM was implemented using a linear kernel.
- SVM is sensitive to feature scales, so standardized data were used to ensure fair distance-based calculations.

Random Forest

- Builds trees independently
- Predicts the final class based on majority voting
- `n_estimators = 200,`
`max_depth = 10,`
`max_features = "sqrt"`
- Model uses Gini impurity as the splitting criterion to select the best feature at each node of the decision trees.

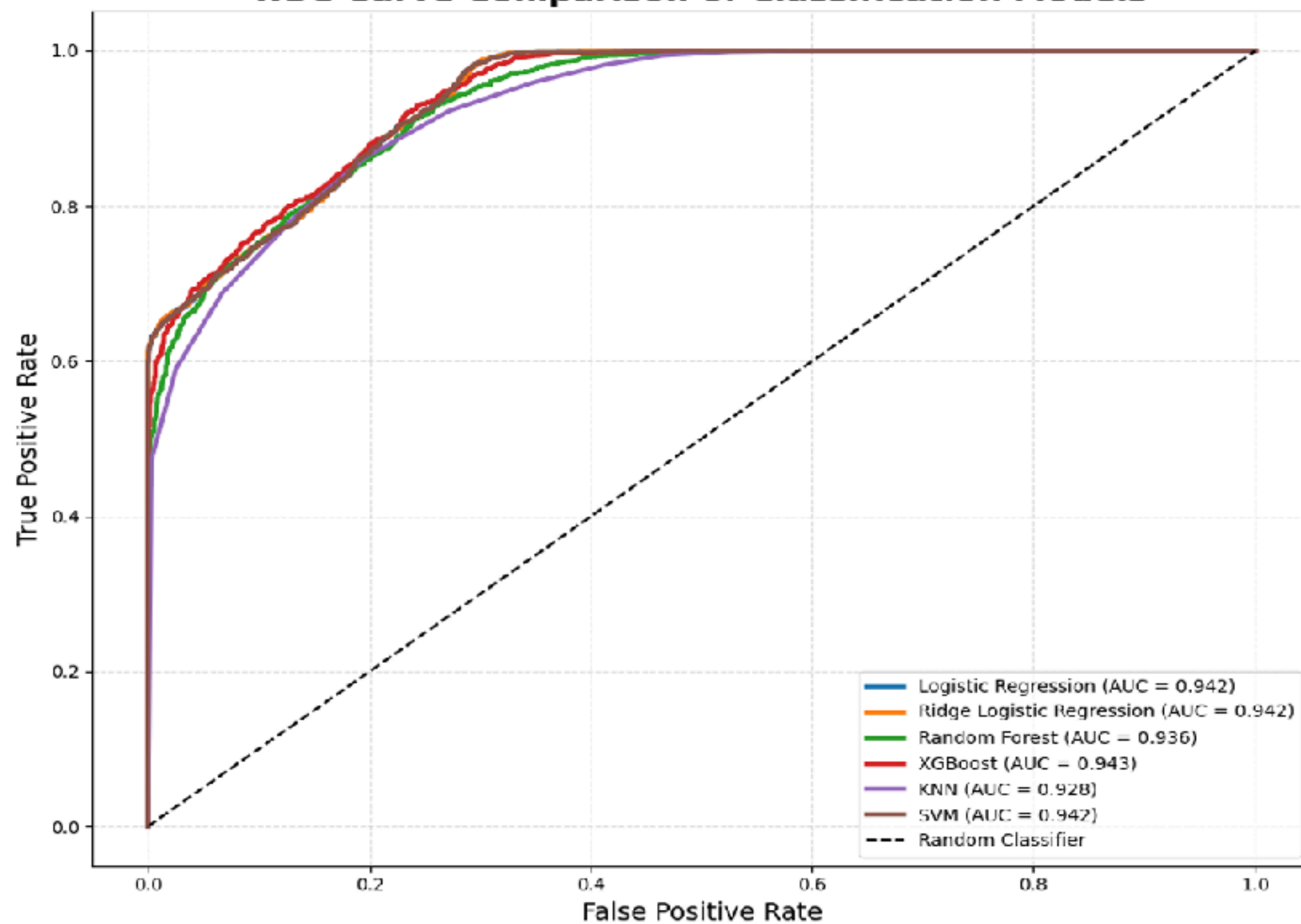
XGBoost

- Builds trees sequentially
- Each new tree corrects the errors made by previous trees
- `n_estimators = 200,`
`max_depth = 3, learning_rate = 0.1`

Note : Optimized hyperparameters for all six models using Scikit-learn's Grid Search CV to achieve the best configuration.

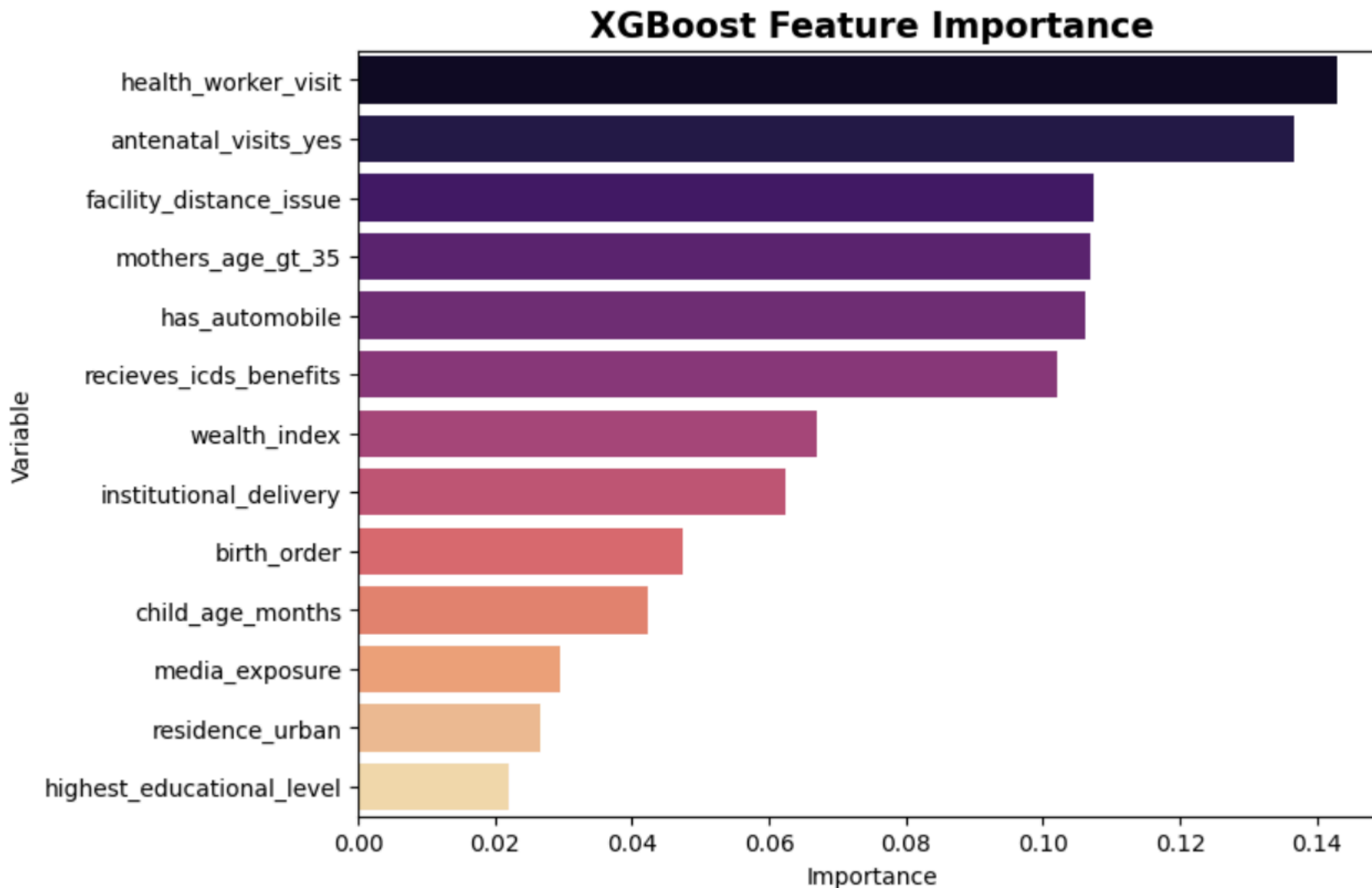
Model	Accuracy	Precision	Sensitivity	Specificity	F1-Score	AUC-ROC
XGBoost ★	86.38%	86.81%	92.44%	76.02%	89.54%	94.27%
Logistic Reg.	85.49%	87.33%	90.05%	77.70%	88.67%	94.21%
Ridge LR	85.38%	87.06%	90.22%	77.11%	88.61%	94.20%
SVM	86.05%	85.88%	93.21%	73.83%	89.39%	94.20%
Random Forest	86.05%	84.55%	95.30%	70.26%	89.60%	93.61%
KNN	85.08%	85.28%	92.27%	72.81%	88.64%	92.84%

ROC Curve Comparison of Classification Models



Confusion Matrix Comparison of Machine Learning Models

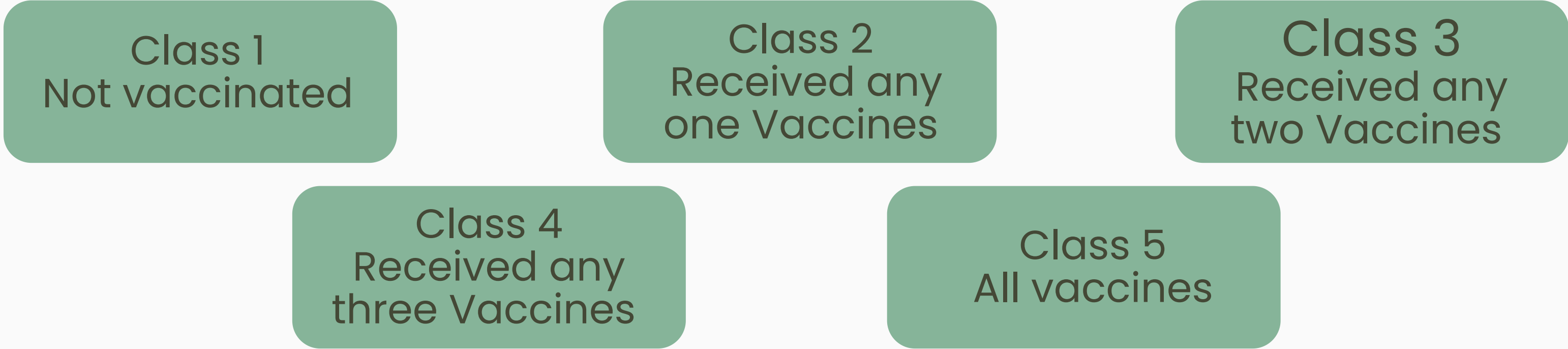
Model		True Positive (TP)	True Negative (TN)	False Positive (FP)	False Negative (FN)
0	Logistic Regression	2109	1066	306	233
1	Ridge Logistic Regression	2113	1058	314	229
2	Random Forest	2232	964	408	110
3	XGBoost	2165	1043	329	177
4	KNN	2161	999	373	181
5	SVM	2183	1013	359	159



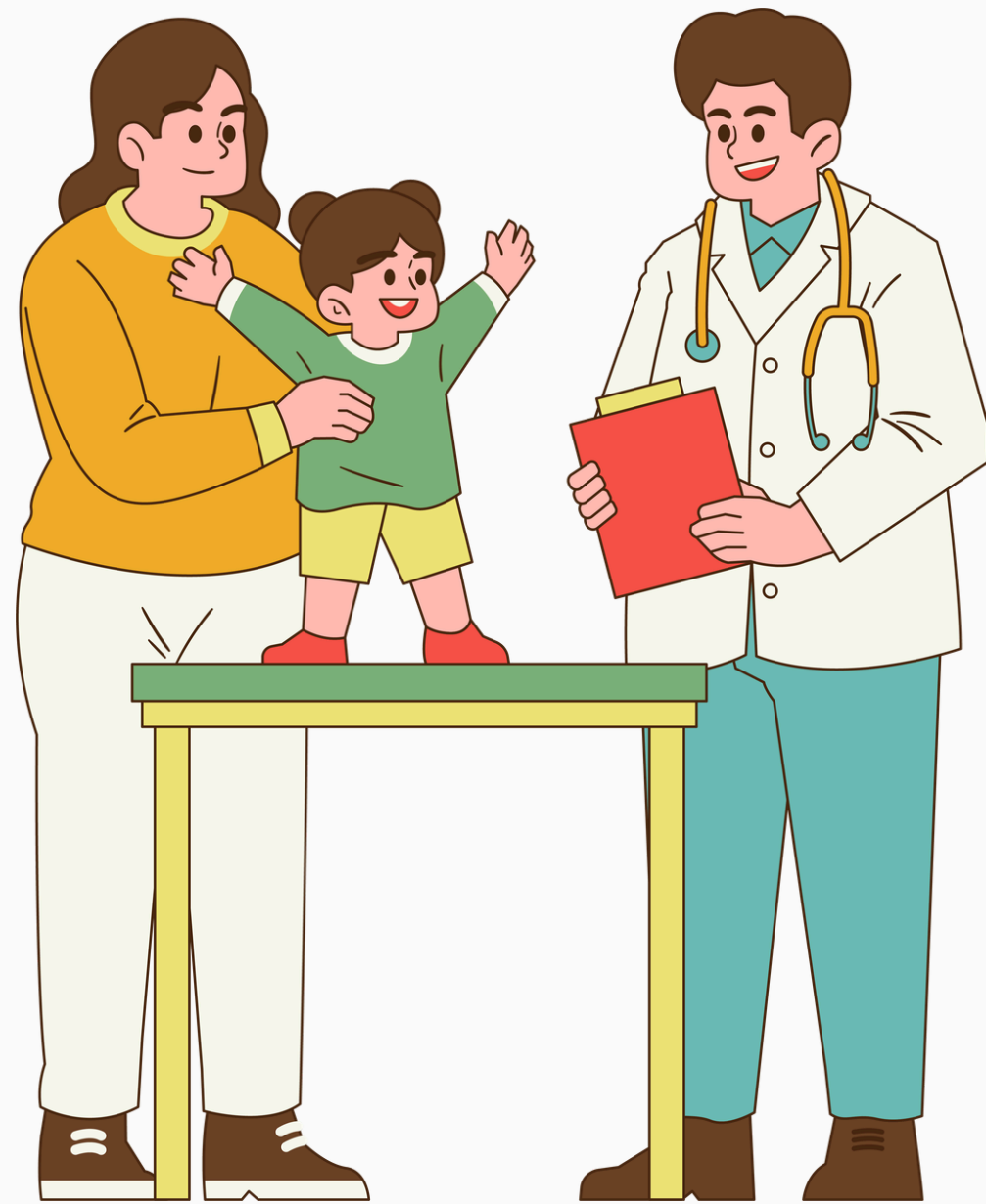
- Child and demographic factors play a smaller role — birth order, child age, media exposure, residence, and maternal education contribute minimally once stronger predictors are accounted for.

- Health-system contact is the strongest predictor — health-worker visits and ANC visits dominate the model, showing that direct engagement with healthcare services drives vaccination uptake.
- Access and socioeconomic factors matter — distance to facility, wealth, institutional delivery, and ICDS benefits show moderate influence, highlighting structural and household-level barriers.

Multiclass vaccination classification



Model	Accuracy	Wtd F1 score	Macro AUC	Micro AUC
Multinomial LR	71.70%	66.36%	84.70%	92.69%
XGBoost	70.25%	66.20%	83.11%	91.84%
Random Forest	69.79%	63.22%	82.63%	91.23%



Study Limitation

- Cross-sectional NFHS-5 data limits causal interpretation.
- Lack of supply-side variables such as facility distance, vaccine stock-outs, and cold chain details.
- Possible recall and social desirability bias in maternally reported vaccination status.
- Random oversampling may increase overfitting due to duplicate samples.
- Survey period overlapped with the COVID-19 pandemic, potentially affecting vaccination patterns.
- National-level analysis may mask important state-level variations.
- Models predict vaccination status retrospectively, limiting real-time intervention use.

Policy Recommendations



- Integrate ML-based risk prediction models into HMIS and RCH portals for real-time identification of high-risk children.
- Strengthen poverty-focused vaccination interventions for economically disadvantaged households.
- Enhance ASHA and ANM workforce capacity to improve outreach and follow-up care.
- Promote institutional deliveries to improve linkage with postnatal immunization services.
- Use antenatal care visits for maternal counseling on childhood vaccination schedules.
- Improve immunization monitoring through better integration of health and administrative databases.

Conclusion

- Machine learning models, especially XGBoost, Random Forest, and Ridge Logistic Regression, achieved high predictive accuracy (AUC > 93%) for childhood vaccination status.
- Key determinants included health worker visits, institutional delivery, maternal education, and household wealth.
- Vaccination coverage declined from early to later vaccines, highlighting the challenge of maintaining healthcare engagement.
- Both demand-side awareness and supply-side healthcare improvements are essential for universal vaccination coverage.
- Interpretable models like Ridge Logistic Regression performed comparably to complex ensemble models, supporting their use in public health policy decisions.

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**THANK YOU
VERY MUCH**